

03_FINAL_ARCHITECTURE.md

SIH26034 - Legal Metrology Automated Compliance System

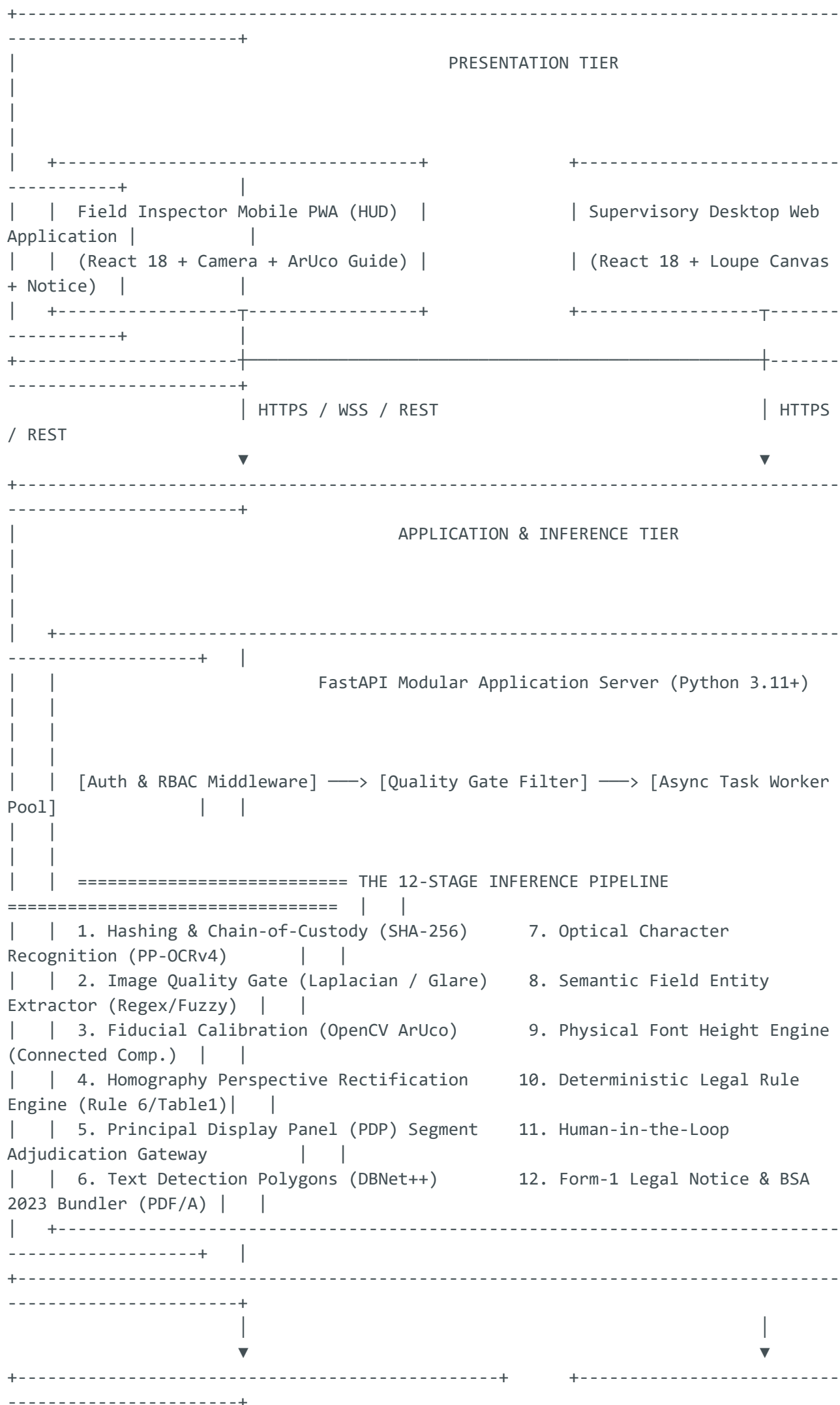
Complete End-to-End System Architecture, 12-Stage Pipeline Flow & Deployment Topology

1. Architectural Philosophy & The "Simplicity Test"

To ensure bulletproof reliability during the Smart India Hackathon 2026 evaluation and prevent distributed systems failure modes under pressure, this system adheres to the **Modular Monolith Architecture**:

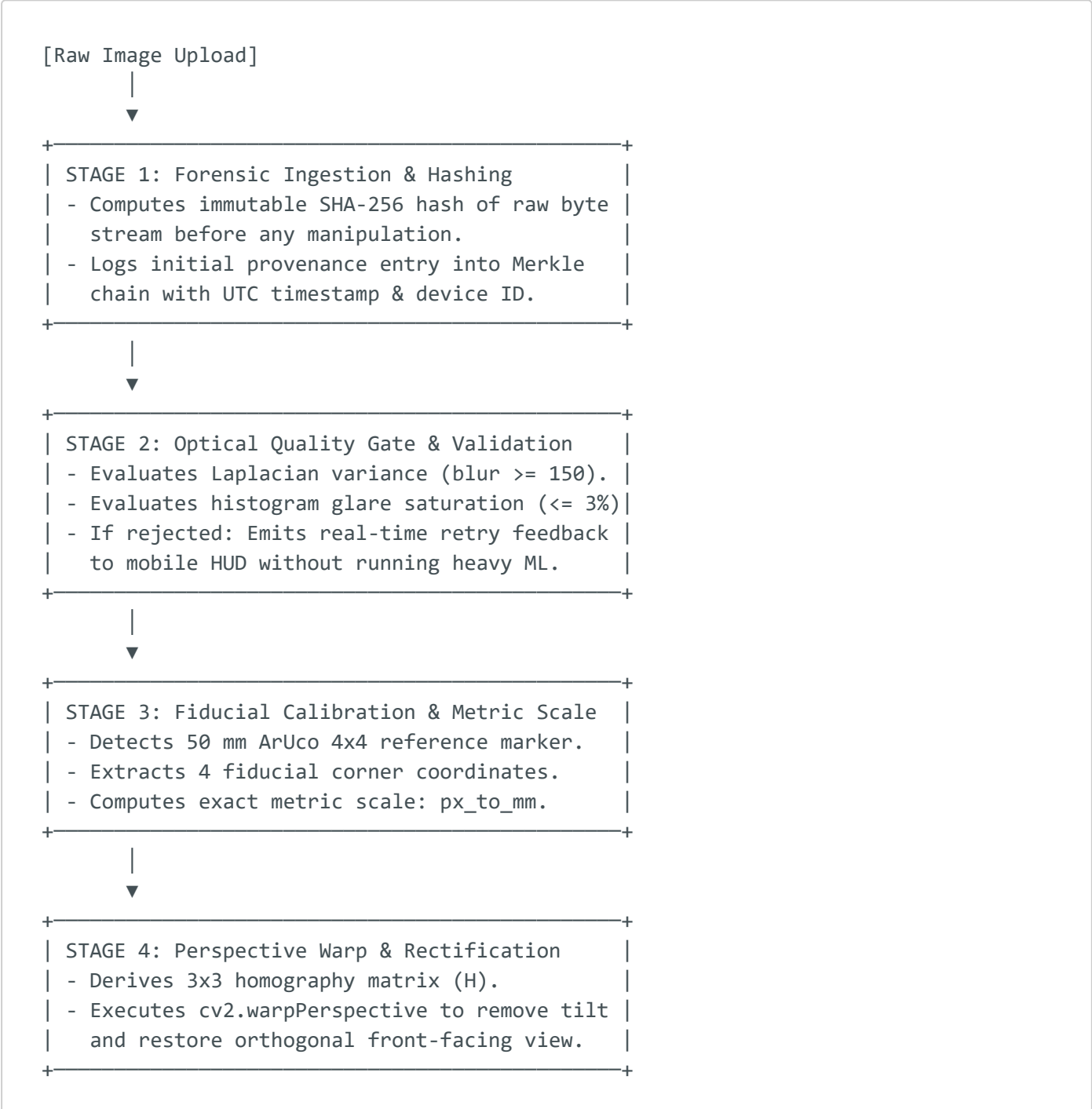
- **Zero Microservices Bloat:** No distributed service meshes, Kafka clusters, or multi-node Kubernetes clusters.
 - **Unified ASGI Process:** A single, high-performance Python 3.11+ FastAPI service orchestrates ingestion, optical pre-processing, ONNX inference, and deterministic rule evaluation.
 - **Embedded Edge Resilience:** Self-contained execution. The system can run entirely disconnected from the cloud on a single developer laptop or field tablet.
 - **Strict Layer Decoupling:** Separation of concerns enforced through clean abstract interfaces, immutable DTOs, and event-driven database logging.
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2. High-Level System Architecture (C4 Level 2 Container Diagram)



	PERSISTENCE TIER	FORENSIC
AUDIT TIER		
- SQLite 3.45+ with SQLCipher (Edge / MVP)		- Append-Only Merkle
Tree Audit Ledger		
- PostgreSQL 16+ with pgcrypto (Cloud)		- SHA-256 Hash Chaining
on all Inspections		
- Local Encrypted File Store (Images & PDFs)		- Section 63 BSA 2023
Digital Certificate Log		
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3. The 12-Stage Inference & Rule Pipeline (Detailed Data Flow)





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| STAGE 5: Principal Display Panel (PDP) Calc |
| - Locates package boundaries via contour/Canny |
| - Calculates total packaging area (cm²). |
| - Calculates PDP area based on geometry: |
|   * Rectangular: 40% of height x width. |
|   * Cylindrical: 40% of height x circumf. |
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| STAGE 6: Multi-Oriented Text Detection |
| - DBNet++ (Real-Time Text Detector, ONNX INT8) |
| - Outputs oriented bounding polygons (4-point) |
|   around all textual clusters on the label. |
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| STAGE 7: Optical Character Recognition (OCR) |
| - PaddleOCR PP-OCRv4 (Permissive Apache-2.0). |
| - Batched line recognition on cropped patches. |
| - Emits raw text strings + character conf. |
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| STAGE 8: Semantic Entity Classification |
| - Fuzzy matching & regex entity parsers: |
|   * MRP & currency tokens (Rs., INR, /-). |
|   * Net Quantity (g, kg, ml, l). |
|   * Unit Sale Price (USP). |
|   * Manufacturer / Packer address keywords. |
|   * Country of Origin ("Made in ..."). |
|   * Dates (Mfg / Expiry / Use by). |
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| STAGE 9: Physical Font Measurement Engine |
| - Connected-Components Analysis on binarized |
|   Net Quantity and MRP text patches. |
| - Isolates lowercase 'x' or numeral heights. |
| - Converts pixel height to millimeters using |
|   calibrated px_to_mm scale. |
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```



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| STAGE 10: Deterministic Legal Metrology Rules |
| - Evaluates Rule 6(1)(h) & Table-I font sizes: |
|   * Compares measured mm vs statutory min mm. |
| - Evaluates Rule 6(1)(k) Unit Sale Price math |
| - Evaluates Rule 6(1)(e) "(incl. of all taxes)" |
| - Evaluates Rule 6(1)(p) Country of Origin. |
| - Emits structured Compliance Evaluation list. |
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```

STAGE 11: Human-in-the-Loop Adjudication Gate

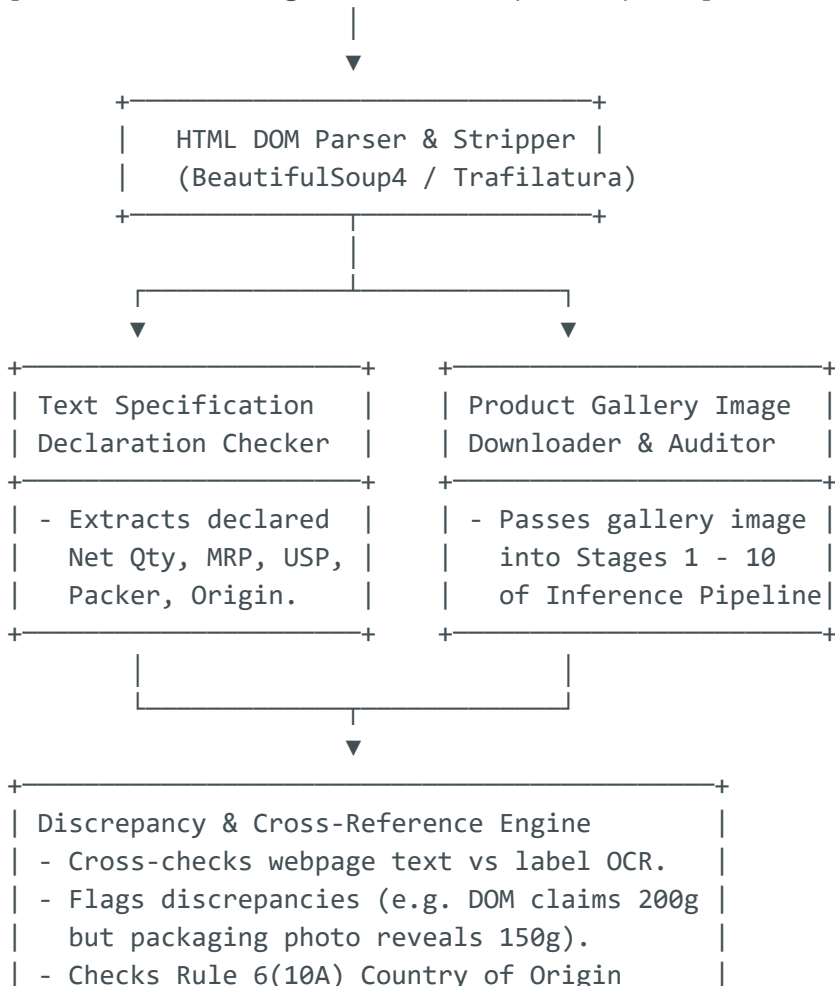
- Renders side-by-side verification canvas.
- Inspector reviews bounding boxes & loupe.
- Officer confirms violation OR overrides AI.
- All overrides require mandatory remarks.

STAGE 12: Evidence Bundler & Notice Generator

- Compiles Form-1 / Form-2 Section 36(1) Statutory Legal Notice.
- Compiles Section 63 BSA 2023 Digital Certificate with cryptographic image hashes.
- Signs and outputs PDF/A evidence bundle.

4. E-Commerce Pipeline Extension (Rule 6(10) & Rule 6(10A))

[E-Commerce Listing URL / DOM Snapshot Upload]



5. Deployment Topology & Operational Readiness

5.1 Local Hackathon Demo Mode (Zero External Infrastructure)

- **Host:** Any standard Intel Core i5/i7 or AMD Ryzen laptop running Windows, macOS, or Linux.
- **Backend:** `uvicorn app.main:app --host 127.0.0.1 --port 8000 --reload`
- **Frontend:** `npm run dev` (Vite dev server running on `localhost:5173`)
- **Database:** Local embedded SQLite with SQLCIPHER (`legal_metrology.db`).
- **Dependencies:** Self-contained within a single Python virtual environment (`.venv`).

5.2 Production State-Wide Cloud Mode (Docker Compose Topology)

```
version: '3.8'
services:
  web-proxy:
    image: nginx:alpine
    ports: ["80:80", "443:443"]
    volumes: [".:/nginx.conf:/etc/nginx/nginx.conf:ro"]
    depends_on: ["frontend", "backend"]

  frontend:
    build: { context: ./frontend, dockerfile: Dockerfile }
    environment: [VITE_API_BASE_URL=/api/v1]

  backend:
    build: { context: ./backend, dockerfile: Dockerfile }
    environment:
      -
DATABASE_URL=postgresql://lmo_admin:secure_pass@postgres:5432/legal_metrology
      - ONNX_NUM_THREADS=4
      - MODEL_CACHE_DIR=/app/models
    volumes: ["evidence_storage:/app/storage"]
    depends_on: ["postgres"]

  postgres:
    image: postgres:16-alpine
    environment:
      - POSTGRES_DB=legal_metrology
```

```
- POSTGRES_USER=lmo_admin
- POSTGRES_PASSWORD=secure_pass
volumes: ["pg_data:/var/lib/postgresql/data"]
```

```
volumes:
  pg_data:
  evidence_storage:
```

6. Resilience & Circuit Breaker Architecture

- 1. Optical Quality Circuit Breaker:** If image blur variance $\sigma^2 < 150$ or glare $> 3\%$, the pipeline aborts immediately before invoking DBNet++ or PP-OCRv4. This prevents wasting CPU cycles on unresolvable images.
- 2. Fallback Calibration Mode:** If the ArUco marker is missing or occluded, the system attempts secondary edge/coin detection. If unresolvable, the system switches to **"Relative / Textual Audit Only"** mode (evaluating presence of declarations and USP math, while gracefully flagging Table-I physical font heights as "Pending Physical Benchmarking").
- 3. Audit Ledger Self-Healing:** The cryptographic Merkle ledger verifies its own chain on startup. If an unauthorized byte modification is detected in the database, the server halts into read-only safety mode and alerts the administrator.